TWO OPEN IMMERSIONS OF FINITE TYPE SEPARATED SCHEMES OVER A COMMON OPEN CAN BE MADE PART OF ONE SEPARATED FINITE TYPE SCHEME BY ADMISSIBLE BLOWUPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project own definitions of the reference, slogan and history display environments (verbatim package plus three newenvironment lines) that the retained file carries at its end and that the project itself typesets as comments. No macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0F3X (sources/stacks-flat/pinned-original-source.tex line 9647).

Allowed standard prerequisites (editorial).

- Scheme theoretic images of immersions and partial sections. Morphisms, Lemmas labelled morphisms-lemma-quasi-compact-immersion and morphisms-lemma-scheme-theoretic-image-of-partial-section, cited at proof lines 9670-9679. Produces X_12 as a closed subscheme of X_1 x_S X_2 with the projections isomorphisms over U.
- Making a scheme theoretic image an open subscheme by an admissible blowup. Same chapter, Lemma labelled lemma-zariski-after-blowup, cited at proof lines 9682-9686; Divisors, Lemmas labelled divisors-lemma-strict-transform, divisors-lemma-blowing-up-two-ideals and divisors-lemma-extend-admissible-blowups, cited at proof lines 9690-9706. Supplies the blowups making the strict transforms open and the blowup of X_12 dominating both.
- Properness of the immersion of the strict transform and closedness of proper images. Morphisms, Lemma labelled morphisms-lemma-image-proper-scheme-closed, cited at proof lines 9702-9706. Concludes that X'_12 -> X_1' x_S X_2' is a closed immersion, so the glueing is legitimate.
- Separatedness of the glued scheme and composition of admissible blowups. Same chapter, Lemma labelled lemma-check-separated, cited at proof lines 9718-9720; Divisors, Lemma labelled divisors-lemma-composition-admissible-blowups, cited at proof lines 9722-9724. Gives separatedness of X over S and closes the proof.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original section 01K2: sources/stacks-chow/tags.txt, source lines 2035–2035. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9667 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 01KH: sources/stacks-chow/tags.txt, source lines 2050–2050. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9668 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01RG: sources/stacks-chow/tags.txt, source lines 2301–2301. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9674 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CNG: sources/stacks-chow/tags.txt, source lines 15891–15891. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9681 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 081S: sources/stacks-flat/pinned-original-source.tex, source lines 9197–9197. Same-chapter pointer into the author environment lemma at source line 9197; invoked at proof line 9685 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080E: sources/stacks-chow/tags.txt, source lines 10179–10179. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9690 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080A: sources/stacks-chow/tags.txt, source lines 10175–10175. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9693 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080M: sources/stacks-chow/tags.txt, source lines 10187–10187. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9704 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01W6: sources/stacks-chow/tags.txt, source lines 2471–2471. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9717 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F3U: sources/stacks-flat/pinned-original-source.tex, source lines 9540–9540. Same-chapter pointer into the author environment lemma at source line 9540; invoked at proof line 9721 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080L: sources/stacks-chow/tags.txt, source lines 10186–10186. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9723 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let $U \to X_1$ and $U \to X_2$ be open immersions of schemes over S and assume U, X_1, X_2 of finite type and separated over S. Then there exists a commutative diagram*

$$\begin{array}{ccccc} X_1' & \longrightarrow & X & \longleftarrow & X_2' \\ \downarrow & \nwarrow & \uparrow & \nearrow & \downarrow \\ X_1 & \longleftarrow & U & \longrightarrow & X_2 \end{array}$$

of schemes over S where $X_i' \to X_i$ is a U-admissible blowup, $X_i' \to X$ is an open immersion, and X is separated and finite type over S.

Proof. Throughout the proof all schemes will be separated of finite type over S. This in particular implies these schemes are quasi-compact and quasi-separated and the morphisms between them are quasi-compact and separated. See Schemes, Sections 01K2 and 01KH. We will use that if $U \to W$ is an immersion of such schemes over S, then the scheme theoretic image Z of U in W is a closed subscheme of W and $U \to Z$ is an open immersion, $U \subset Z$ is scheme theoretically dense, and $U \subset Z$ is dense topologically. See Morphisms, Lemma 01RG.

Let $X_{12} \subset X_1 \times_S X_2$ be the scheme theoretic image of $U \to X_1 \times_S X_2$. The projections $p_i : X_{12} \to X_i$ induce isomorphisms $p_i^{-1}(U) \to U$ by Morphisms, Lemma 0CNG. Choose a U-admissible blowup $X_i^i \to X_i$ such that the strict transform X_{12}^i of X_{12} is isomorphic to an open subscheme of X_i^i, see Lemma 081S. Let $\mathcal{I}_i \subset \mathcal{O}_{X_i}$ be the corresponding finite type quasi-coherent sheaf of ideals. Recall that $X_{12}^i \to X_{12}$ is the blowup in $p_i^{-1}\mathcal{I}_i\mathcal{O}_{X_{12}}$, see Divisors, Lemma 080E. Let X_{12}' be the blowup of X_{12} in $p_1^{-1}\mathcal{I}_1 p_2^{-1}\mathcal{I}_2\mathcal{O}_{X_{12}}$, see Divisors, Lemma 080A for what this entails. We obtain in particular a commutative diagram

$$\begin{array}{ccc} X_{12}' & \longrightarrow & X_{12}^2 \\ \downarrow & & \downarrow \\ X_{12}^1 & \longrightarrow & X_{12} \end{array}$$

where all the morphisms are U-admissible blowing ups. Since $X_{12}^i \subset X_i^i$ is an open we may choose a U-admissible blowup $X_i' \to X_i^i$ restricting to $X_{12}' \to X_{12}^i$, see Divisors, Lemma 080M. Then $X_{12}' \subset X_i'$ is an open subscheme and the diagram

$$\begin{array}{ccc} X_{12}' & \longrightarrow & X_i' \\ \downarrow & & \downarrow \\ X_{12}^i & \longrightarrow & X_i^i \end{array}$$

is commutative with vertical arrows blowing ups and horizontal arrows open immersions. Note that $X_{12}' \to X_1' \times_S X_2'$ is an immersion and proper (use that $X_{12}' \to X_{12}$ is proper and $X_{12} \to X_1 \times_S X_2$ is closed and $X_1' \times_S X_2' \to X_1 \times_S X_2$ is separated and apply Morphisms, Lemma 01W6). Thus $X_{12}' \to X_1' \times_S X_2'$ is a closed immersion. It follows that if we define X by glueing X_1' and X_2' along the common open subscheme X_{12}', then $X \to S$ is of finite type and separated (Lemma 0F3U). As

compositions of U-admissible blowups are U-admissible blowups (Divisors, Lemma 080L) the lemma is proved. $\square$